An Arbitrage-Aware SPX Volatility Surface

Identification, Constrained SVI, Heston Calibration, and Honest Validation

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Abstract

This study constructs an auditable volatility surface from a delayed public SPX option-chain snapshot. Forward prices and discount factors are identified by robust put–call parity, implied volatilities are obtained by no-arbitrage inversion, and each maturity is fitted with a constrained raw-SVI smile. A five-parameter Heston model is then estimated by multi-start nonlinear least squares on a deterministic training set and judged on strikes excluded before calibration. The held-out IV-equivalent root-mean-square error is 0.01464, versus 0.04459 for a maturity-level flat-volatility null. This improvement is economically substantial but not a certificate of truth: quotes are asynchronous delayed transactions, one parity fit touches its economically imposed discount bound, and the unconstrained best Heston fit violates the Feller sufficient condition. Those weaknesses are reported as results rather than hidden as implementation details.

What this paper can defend

The repository can reproduce every table and figure from an immutable raw snapshot; the surface satisfies explicit dense-grid static-arbitrage diagnostics; the Heston comparison uses data not seen during fitting; and numerical uncertainty, multi-start stability, data limitations, and failed structural restrictions are visible.

1 Research question and falsification design

The question is narrow: *does a single stochastic-volatility parameter vector explain the cross-section of SPX smiles better than a transparent flat-volatility benchmark, without allowing the preliminary surface fit to manufacture static arbitrage?* The following statements were fixed in code before calibration.

1. The observable information set ends at the common reference session.
2. Four expirations are selected near 30, 90, 180, and 365 calendar days, with standard monthly contracts included in the candidate pool.
3. Every fourth strike within each selected calibration slice is held out deterministically. Heston parameters never see those observations.
4. The null assigns each maturity the liquidity-weighted training-sample mean implied volatility.
5. Failure is possible: negative butterfly density, calendar inversion, unstable optimization, implausible parity, or no held-out improvement would all weaken the model.

The design is a cross-sectional specification test, not a forecast. The option prices are risk-neutral objects and do not estimate physical return probabilities.

2 Contracts, observations, and information time

SPX options are used because their European exercise convention makes the put–call-parity equality coherent. The underlying and chains came from a delayed public Yahoo Finance interface through `yfinance`. It is a convenient research feed, not an exchange-certified historical record. The raw response, ingestion timestamp, library version, candidate-expiration audit, and SHA-256 manifest are committed with the analysis.

Displayed bid and ask fields were too sparse to define a usable common cross-section. The study therefore uses the latest reported transaction for each contract whose timestamp falls within the reference session (with a short post-close reporting allowance). Call and put transactions at a common strike are generally asynchronous. Liquidity weights and the call–put timestamp gap enter the parity fit, but this cannot eliminate staleness.

Assumption 1 (Measurement equation)**.** *For contract i, the observed transaction price satisfies*

$$P_i^{\text{obs}} = P_i^* + \varepsilon_i,$$

where P_i^* is a synchronous frictionless value and ε_i includes bid–ask location, reporting delay, discreteness, and stale-price error. The pipeline propagates a parity-residual scale as an empirical noise benchmark; it does not interpret that scale as an independent Gaussian standard error.

3 Forward and discount identification

For European calls and puts with common strike K and maturity T,

$$C(K,T) - P(K,T) = D(T)\,[F(T) - K] = \alpha_T - \beta_T K, \qquad \alpha_T = D(T)F(T), \quad \beta_T = D(T). \tag{1}$$

Thus a cross-section of paired strikes identifies the slope $-D(T)$ and intercept $D(T)F(T)$. Estimation uses a soft-L^1 loss and liquidity weights. To prevent an asynchronous public chain from implying economically nonsensical discounting, $D(T)$ is bounded by continuously compounded rates between zero and 15 percent.

Proposition 1 (Parity recovery)**.** *If equation (1) holds at two distinct strikes and $D(T) > 0$, then $D(T)$ and $F(T)$ are uniquely identified.*

Proof. Subtract parity at K_1 and K_2. Since $K_1 \neq K_2$,

$$D(T) = -\frac{[C_1 - P_1] - [C_2 - P_2]}{K_1 - K_2}.$$

Substitution into either equation gives $F(T) = K_i + [C_i - P_i]/D(T)$. Uniqueness follows from the nonzero strike difference. □

Table 1: Robust put–call-parity estimates. Prices and strikes are index points.

expiration	maturity_years	forward	discount	parity_rmse	parity_pairs	discount_at_upper_bound
2026-09-18	0.0602	7740.8471	0.9978	4.9360	7	False
2026-11-20	0.2327	7784.2754	1.0000	11.1077	7	True
2027-02-19	0.4819	7872.5407	0.9667	1.9225	5	False
2027-08-20	0.9802	8011.1433	0.9486	6.0432	10	False

Identification warning

The November discount estimate touches its upper bound $D = 1$. The fitted forward is usable as a regularized nuisance estimate, but the corresponding zero rate is not treated as an economic discovery. This boundary event is preserved in the diagnostics and provenance.

4 Black inversion and total variance

With forward F, discount D, volatility σ, and $d_{1,2} = [\log(F/K) \pm \frac{1}{2}\sigma^2 T]/(\sigma\sqrt{T})$, Black's forward formula is

$$C = D[F\Phi(d_1) - K\Phi(d_2)], \qquad P = D[K\Phi(-d_2) - F\Phi(-d_1)].$$

Only out-of-the-money options are retained: puts for $K < F$ and calls for $K \geq F$. The implied volatility is the unique root when the observed price lies strictly inside the no-arbitrage bounds. Root finding uses a bracketed Brent solver, avoiding the vega instability of an unconstrained Newton step.

Log-forward moneyness and total implied variance are

$$k = \log(K/F), \qquad w(k, T) = \sigma_{\mathrm{imp}}(k, T)^2 T.$$

Total variance, rather than volatility, is the natural object for both SVI fitting and calendar-spread diagnostics.

5 Constrained SVI surface

Each maturity uses the raw-SVI parameterization

$$w(k) = a + b\left[\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\right], \tag{2}$$

with $b \geq 0$, $|\rho| < 1$, $\sigma > 0$, and $a + b\sigma\sqrt{1 - \rho^2} \geq 0$. The fit minimizes a liquidity-weighted squared error in total variance. Numerical constraints are checked on a grid wider than the observed strikes, not only at quoted points.

For a twice differentiable total-variance smile, the normalized state-price-density factor is

$$g(k) = \left(1 - \frac{kw'(k)}{2w(k)}\right)^2 - \frac{[w'(k)]^2}{4}\left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{1}{2}w''(k). \tag{3}$$

A nonnegative g is the relevant butterfly-arbitrage diagnostic under the standard SVI transformation. The implementation also enforces finite Lee-wing slopes and tests $w(k, T_{j+1}) - w(k, T_j) \geq 0$ on a common grid. If needed, the later slice receives the smallest vertical total-variance shift that repairs the grid; no such shift was needed in this snapshot.

Table 2: SVI errors and dense-grid arbitrage diagnostics.

expiration	quotes	iv_rmse	iv_mae	inside_empirical_noise_fraction	minimum_total_variance	minimum_butterfly_g	minimum_calendar_variance_gap	left_wing_slope	right_wing_slope
2026-09-18	152	0.00446	0.00363	0.90789	0.00062	0.00000	NaN	0.05433	0.05015
2026-11-20	115	0.00215	0.00157	0.98261	0.00314	0.05997	0.00026	0.09928	0.05787
2027-02-19	64	0.00147	0.00109	0.75000	0.00725	0.16569	0.00059	0.12893	0.05159
2027-08-20	60	0.00126	0.00097	0.80000	0.01738	0.31977	0.00572	0.16300	0.05163

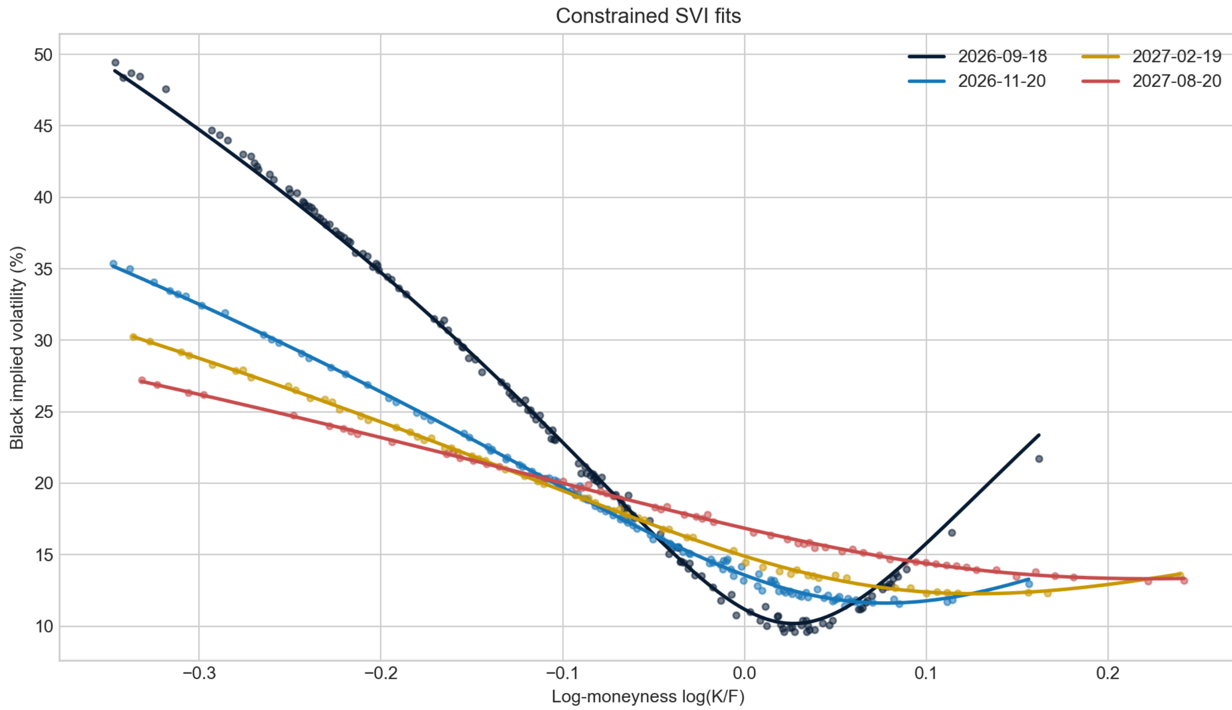

Figure 1: Observed implied volatilities and constrained SVI fits.

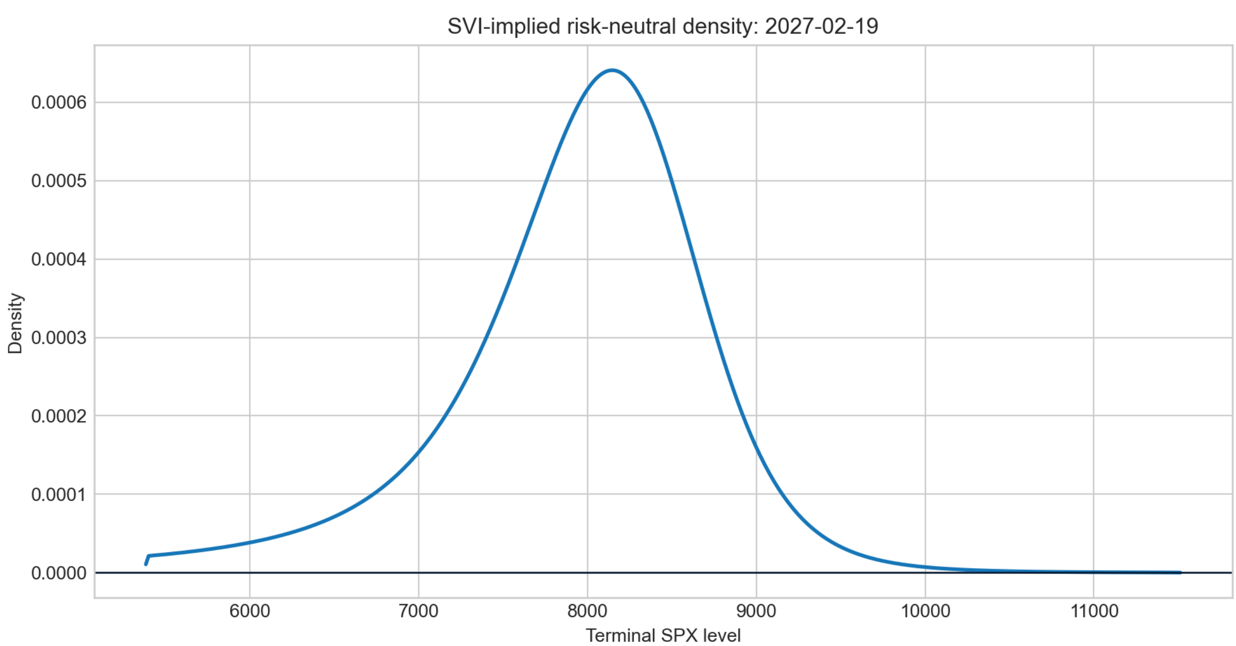

Figure 2: State-price-density diagnostic implied by equation (3).

6 Heston structural compression

Under the pricing measure, the Heston state follows

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sqrt{v_t}S_t\,\mathrm{d}W_t^S, \tag{4}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t. \tag{5}$$

The parameters have distinct interpretations: κ is mean-reversion speed, θ long-run variance, ξ volatility of variance, ρ the return–variance correlation, and v_0 initial variance. Calls are evaluated by Fourier inversion of the characteristic function. Puts follow from parity.

Calibration minimizes price residuals normalized by Black vega,

$$\min_{\vartheta} \sum_{i\in\mathcal{T}} \left[\frac{P_i^{\mathrm{Heston}}(\vartheta) - P_i^{\mathrm{obs}}}{\max(\mathrm{Vega}_i, \epsilon)}\right]^2,$$

so the objective is approximately an implied-volatility error rather than a currency-weighted error dominated by expensive contracts. Five dispersed starting points test local-solution sensitivity. Approximate standard errors use the local least-squares Jacobian and should be read as curvature diagnostics, not as sampling theory for dependent noisy quotes.

Table 3: Heston calibration and local curvature uncertainty.

parameter	estimate	local_standard_error	lower_bound	upper_bound	near_boundary	selected_start
kappa	2.84591	0.99035	0.05000	12.00000	False	5
theta	0.05587	0.00866	0.00500	0.60000	False	5
vol_of_vol	1.10912	0.10695	0.05000	3.00000	False	5
rho	-0.67985	0.02961	-0.99000	0.99000	False	5
initial_variance	0.01682	0.00153	0.00500	0.60000	False	5

Feller condition

The best fit has $2\kappa\theta - \xi^2 < 0$. The variance process remains definable, but strict positivity is not guaranteed by the Feller sufficient condition. Imposing that condition would change the empirical question and should be reported as a separate restricted specification, not silently forced into the baseline.

7 Held-out comparison

Table 4: Training and held-out errors. The holdout was excluded before calibration.

model	sample	quotes	price_rmse	iv_equivalent_rmse	iv_equivalent_mae	inside_empirical_noise_fraction
flat	holdout	24	50.28291	0.04459	0.03827	0.08333
heston	holdout	24	5.44677	0.01464	0.00903	0.58333
flat	train	76	49.91876	0.04116	0.03606	0.18421
heston	train	76	4.32688	0.01421	0.00812	0.59211

On 24 held-out contracts, Heston reduces IV-equivalent RMSE from 0.04459 to 0.01464 and price RMSE from 50.28 to 5.45 index points. It places 58.3 percent of holdout residuals inside the empirical noise band, versus 8.3 percent for the null. The result rejects the claim that a maturity-level flat volatility is an adequate cross-sectional description of this snapshot. It does *not* establish that Heston is the true data-generating process.

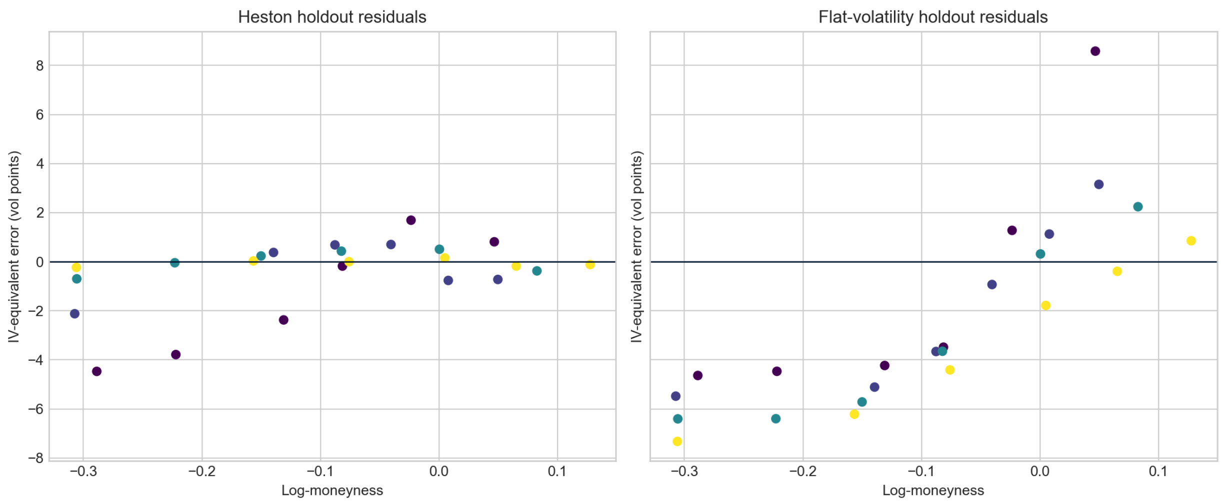

Figure 3: Held-out IV-equivalent residuals by log-forward moneyness.

8 Model-risk register

Risk	Observable control	Remaining limitation
Delayed transactions	Exact contract timestamps, common-session filter, immutable raw file	Prices are asynchronous and not a synchronized NBBO.
Parity identification	Robust loss, liquidity weights, rate bounds, boundary flag	Sparse call–put overlap can still bias forward and discount estimates.
Static arbitrage	Black bounds, dense SVI $g(k)$ test, Lee wings, calendar grid	Finite-grid tests are numerical certificates on a declared domain, not global proofs.
Optimization	Multiple Heston starts and retained start-level table	Agreement across starts does not eliminate common numerical or specification error.
Overfitting	Strike holdout and explicit flat-volatility null	One date and deterministic split do not establish temporal stability.
Structural validity	Feller slack and local Jacobian uncertainty reported	The best unrestricted fit violates the Feller sufficient condition.

9 Reproduction and audit trail

From the repository root, run

```
python projects/volatility_surface/run_research.py
python scripts/execute_notebooks.py notebooks/07_spx_surface_heston_validation.ipynb
powershell -ExecutionPolicy Bypass -File projects/volatility_surface/build_report.ps1
```

The default research run uses the committed snapshot and is network independent. Passing `-refresh` explicitly requests a new public-chain download. Each generated file is covered by `output/manifest.sha256`; the provenance record states whether the run used a refresh or the snapshot.

10 Conclusion

The strongest result is methodological, not ornamental. A defensible volatility surface requires a chain of controls: contract convention, information timestamps, parity identification, no-arbitrage price inversion, constrained smile geometry, structural calibration, untouched observations, a meaningful null, and explicit failure modes. In this snapshot the Heston compression is materially better than flat volatility, while its negative Feller slack and the parity boundary identify exactly where the structural story should not be oversold.

References

[1] J. Gatheral and A. Jacquier, *Arbitrage-free SVI volatility surfaces*, Quantitative Finance 14(1), 2014. https://arxiv.org/abs/1204.0646

[2] S. L. Heston, *A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options*, Review of Financial Studies 6(2), 1993. https://doi.org/10.1093/rfs/6.2.327

[3] Options Industry Council, *Understanding the Bid and Ask Prices for Options*. https://www.optionseducation.org/news/understanding-the-bid-and-ask-prices-for-options